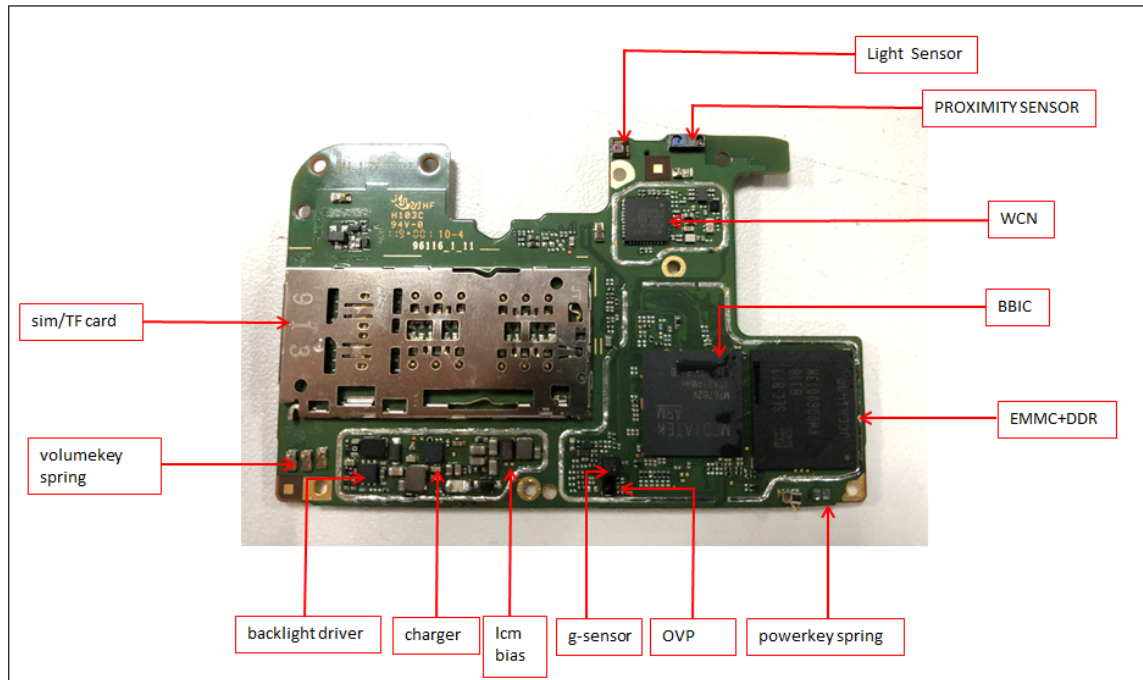


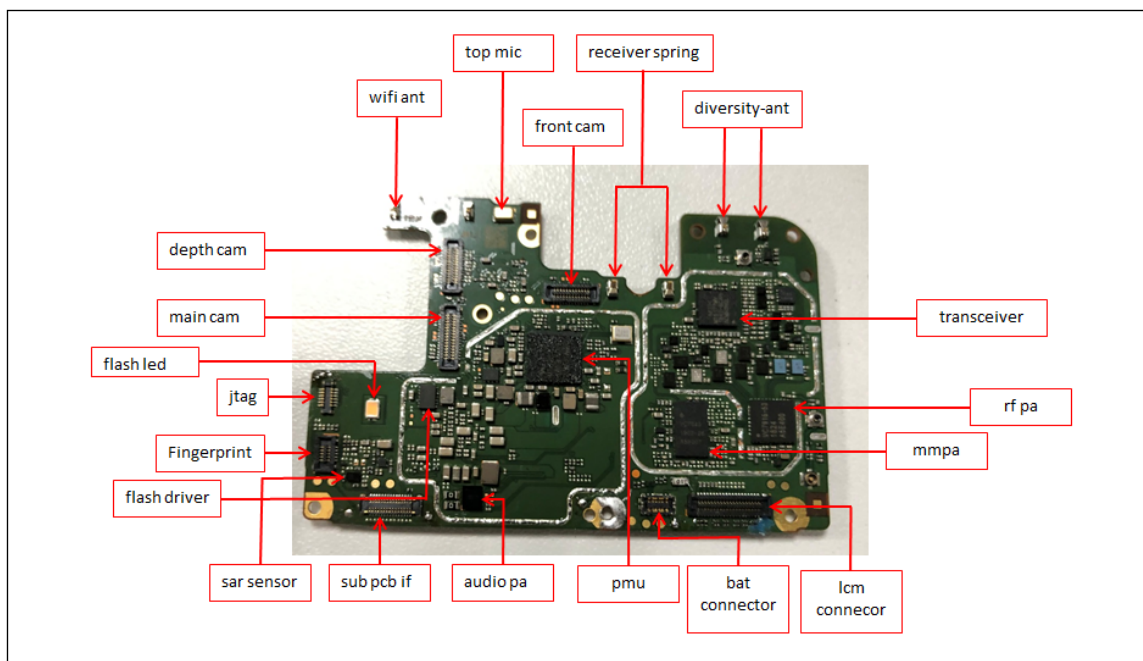
8. Level 3 Repair

8-1. Components Layout

Main board top view

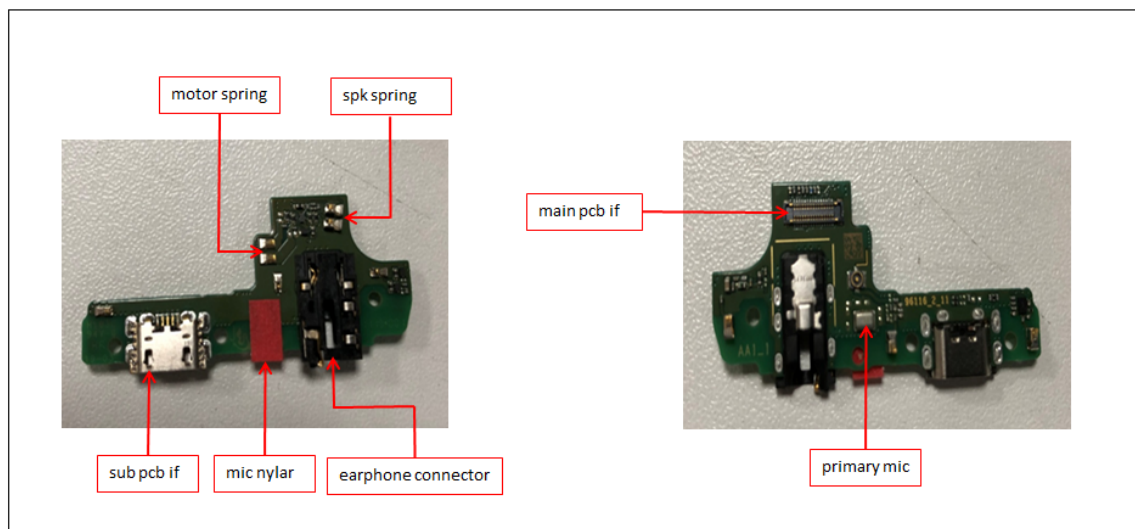


Main board bot view

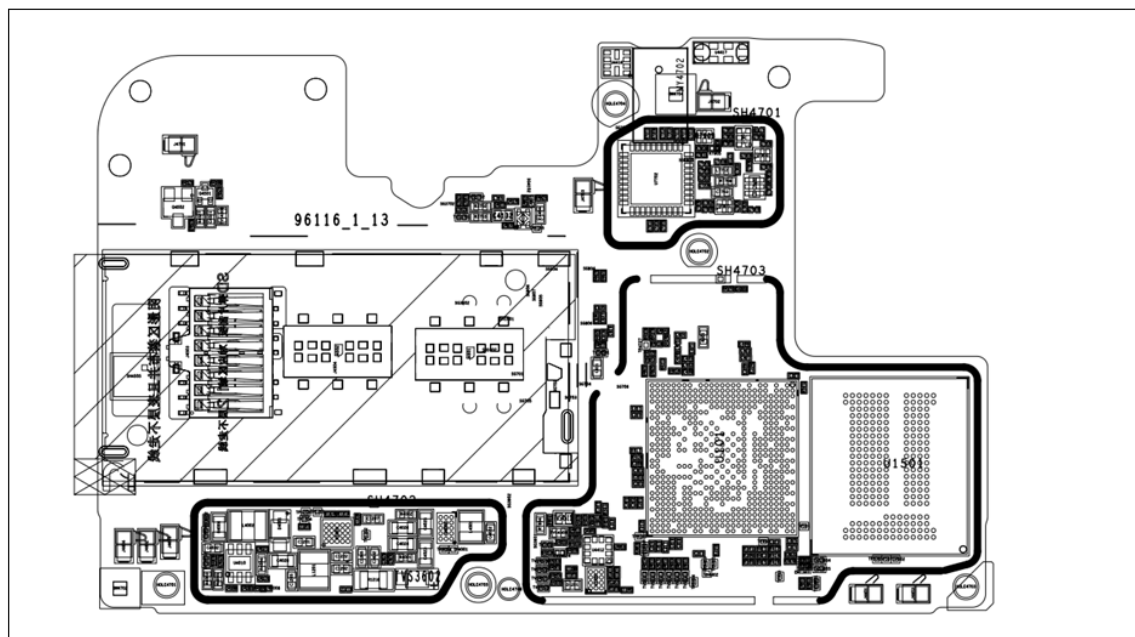


8. Level 3 Repair

Sub board

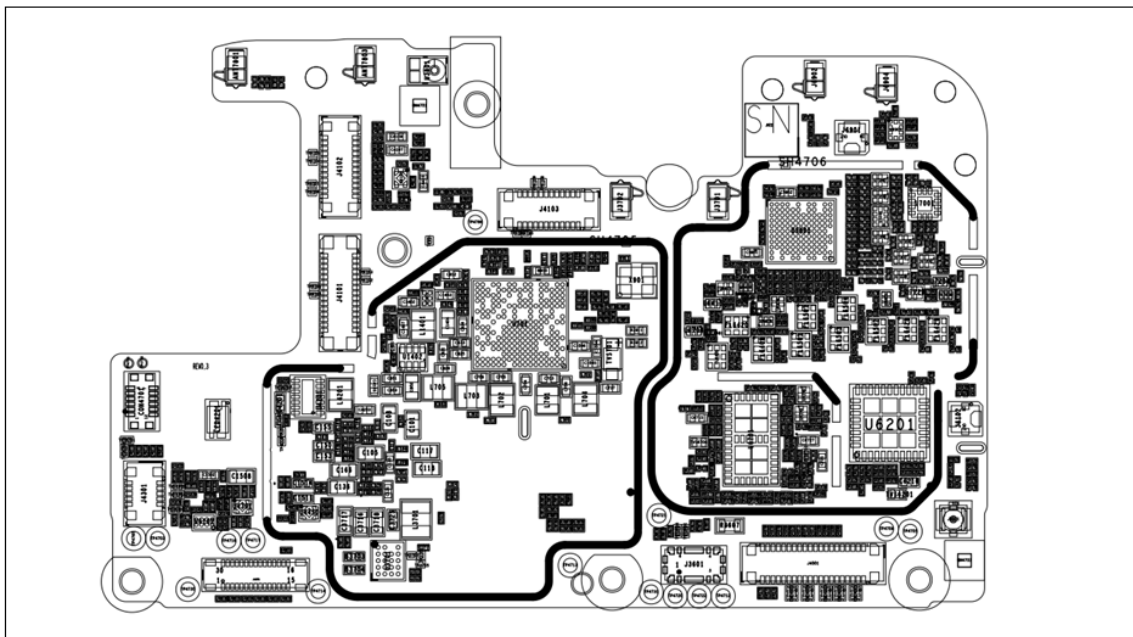


Reference map Mb top view

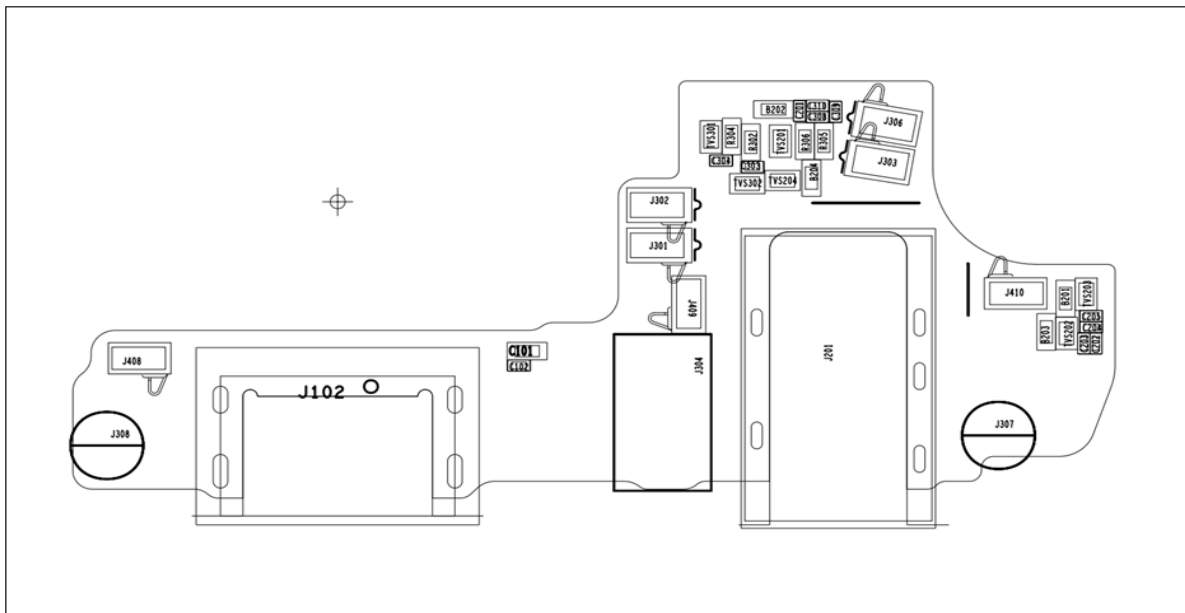


8. Level 3 Repair

Mb bot view

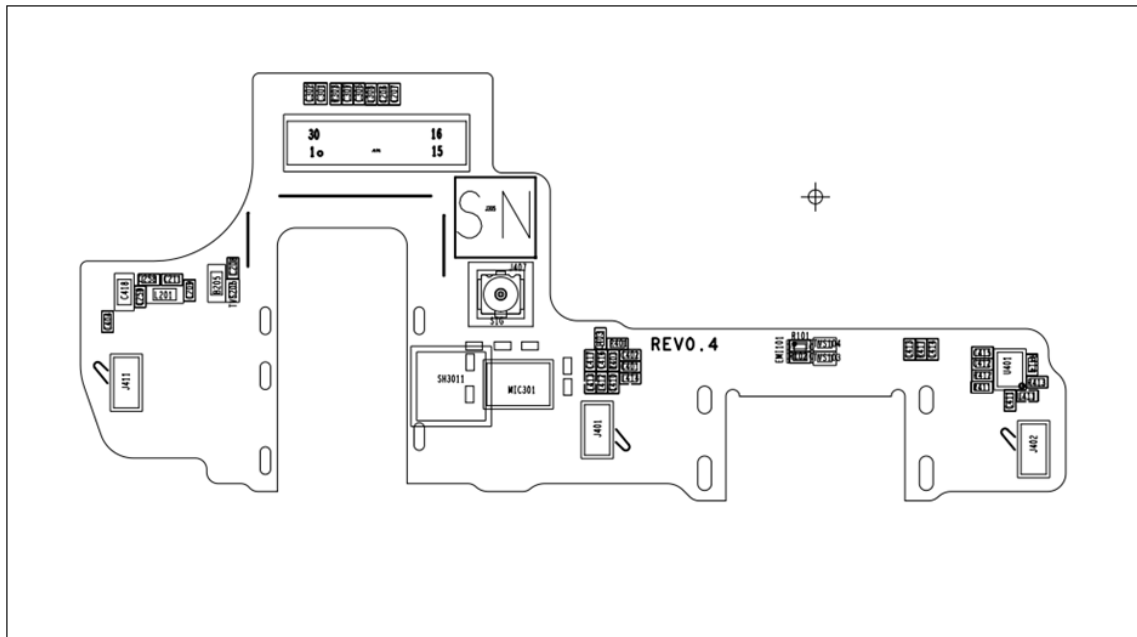


Sub top view

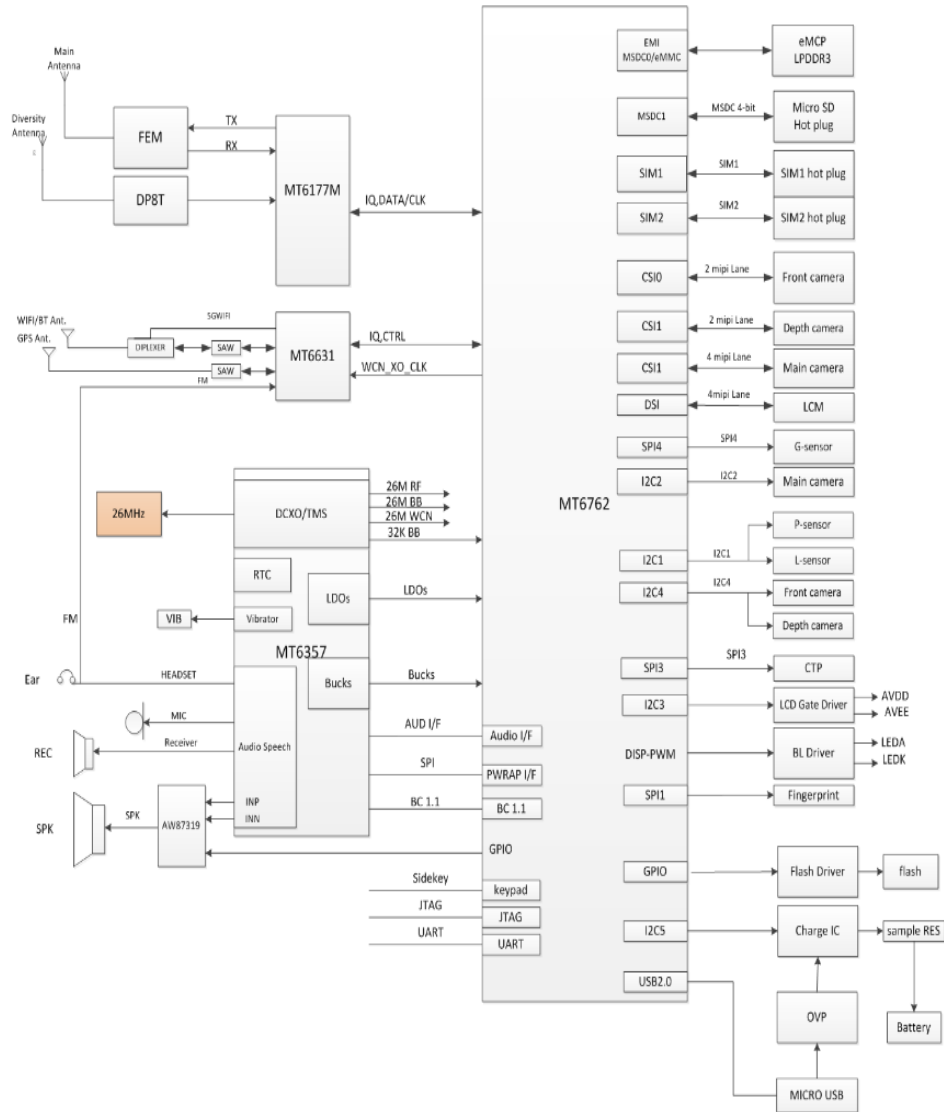


8. Level 3 Repair

Sub bot view



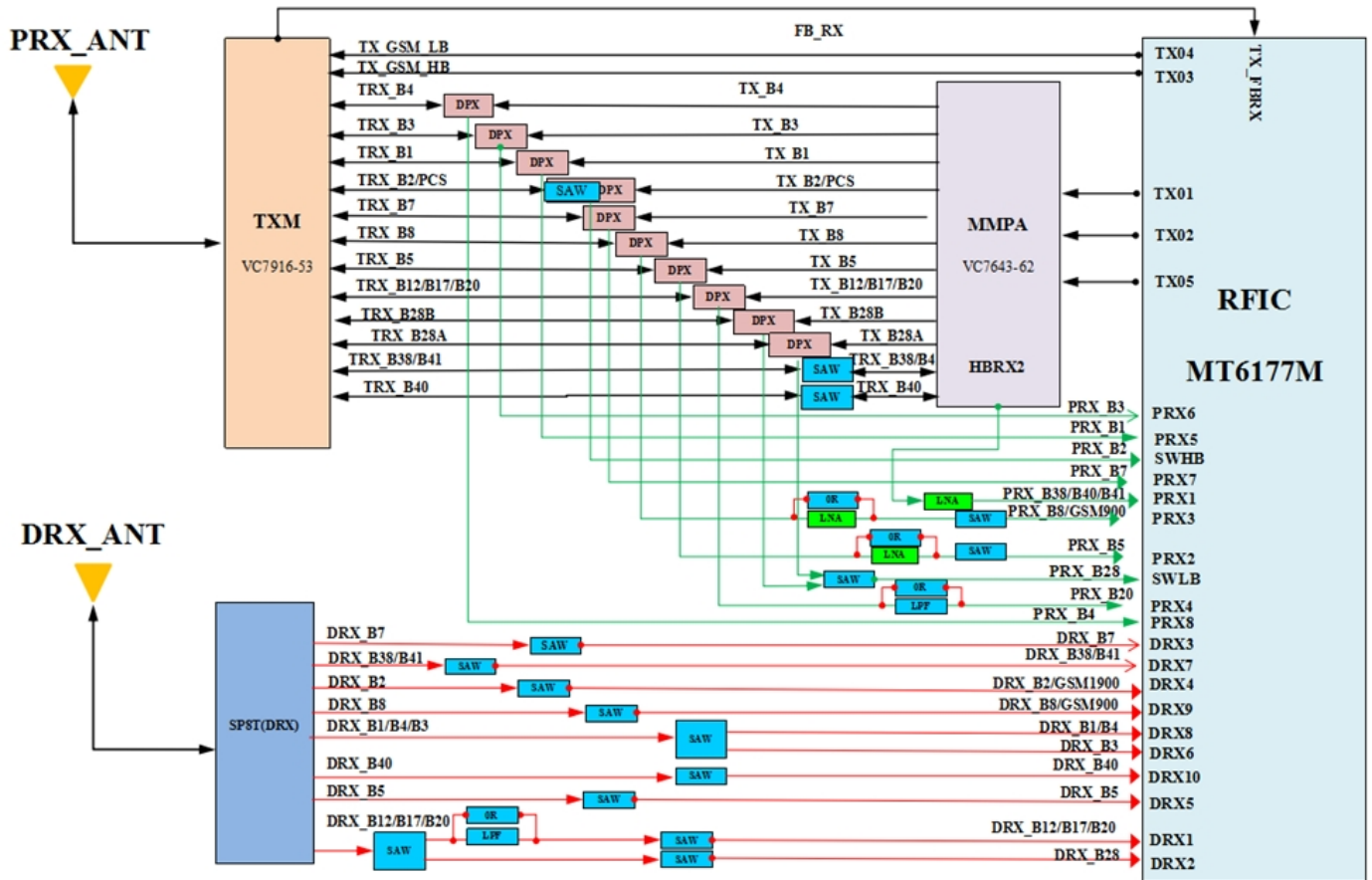
8. Level 3 Repair



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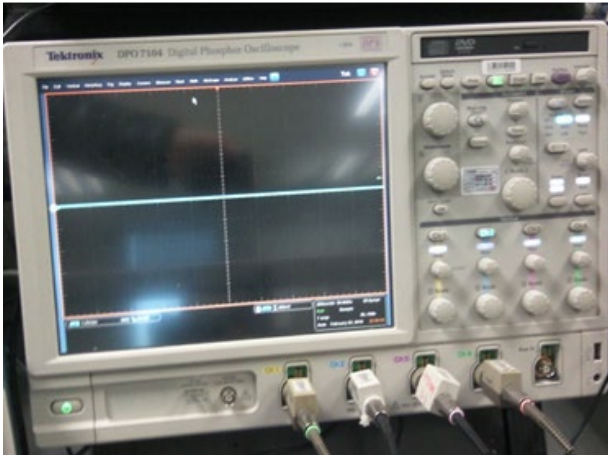
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8. Level 3 Repair



8. Level 3 Repair

8-3. Flow chart of Troubleshooting.



Oscilloscope



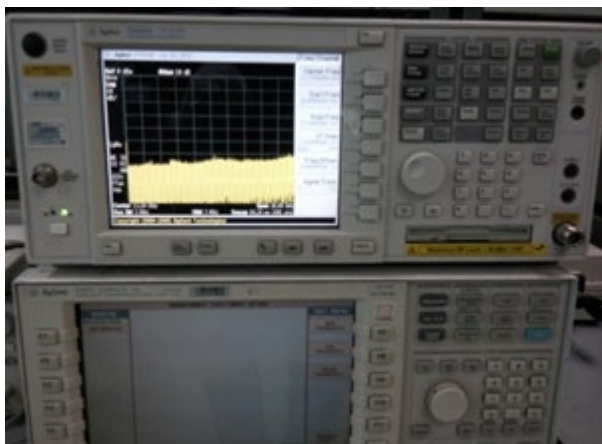
Digital multimeter



Power Supply



+ driver, ESD Safe Tweezer



8960 & Spectrum Analyzer

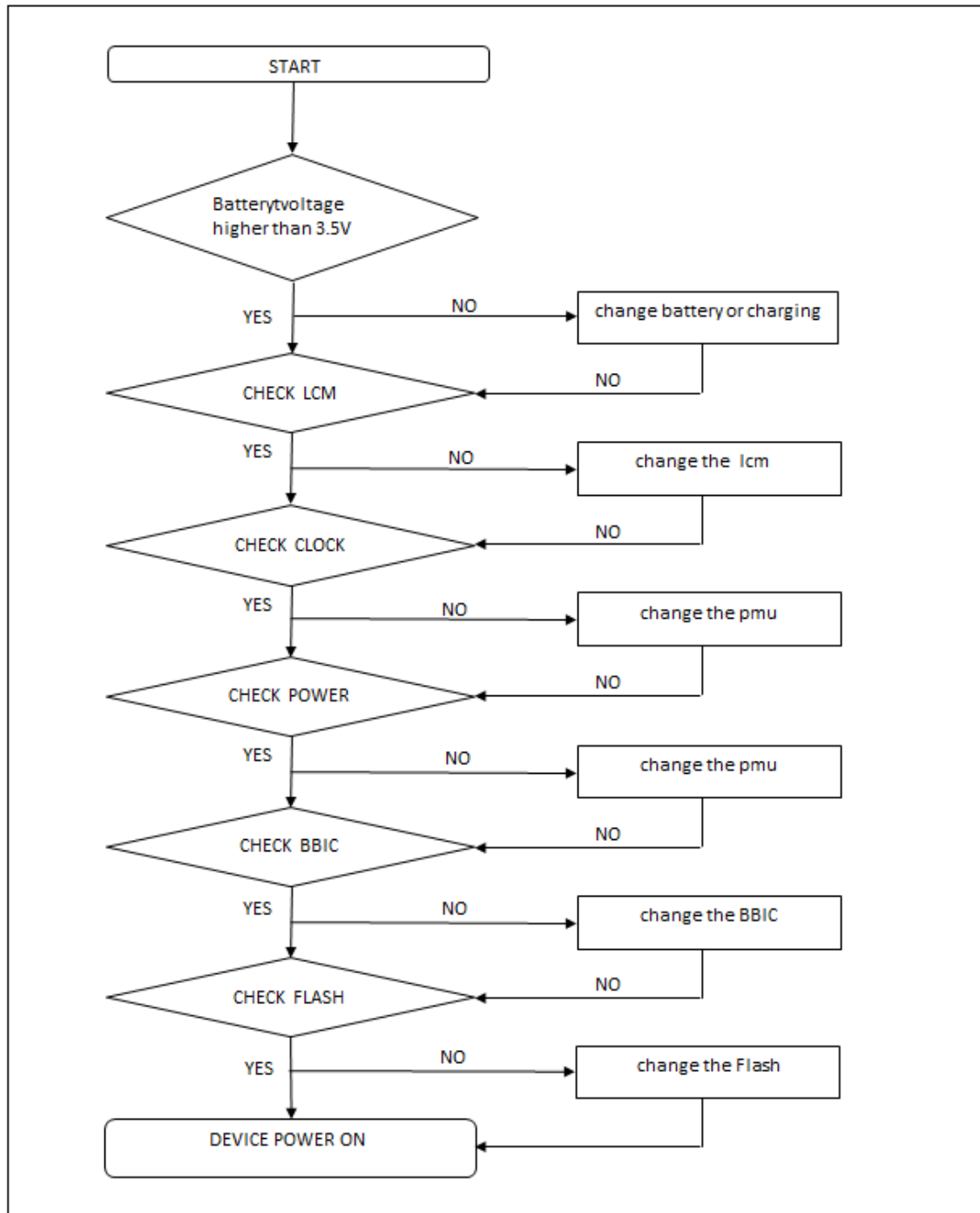


Soldering iron

8. Level 3 Repair

8-4-1. Power On

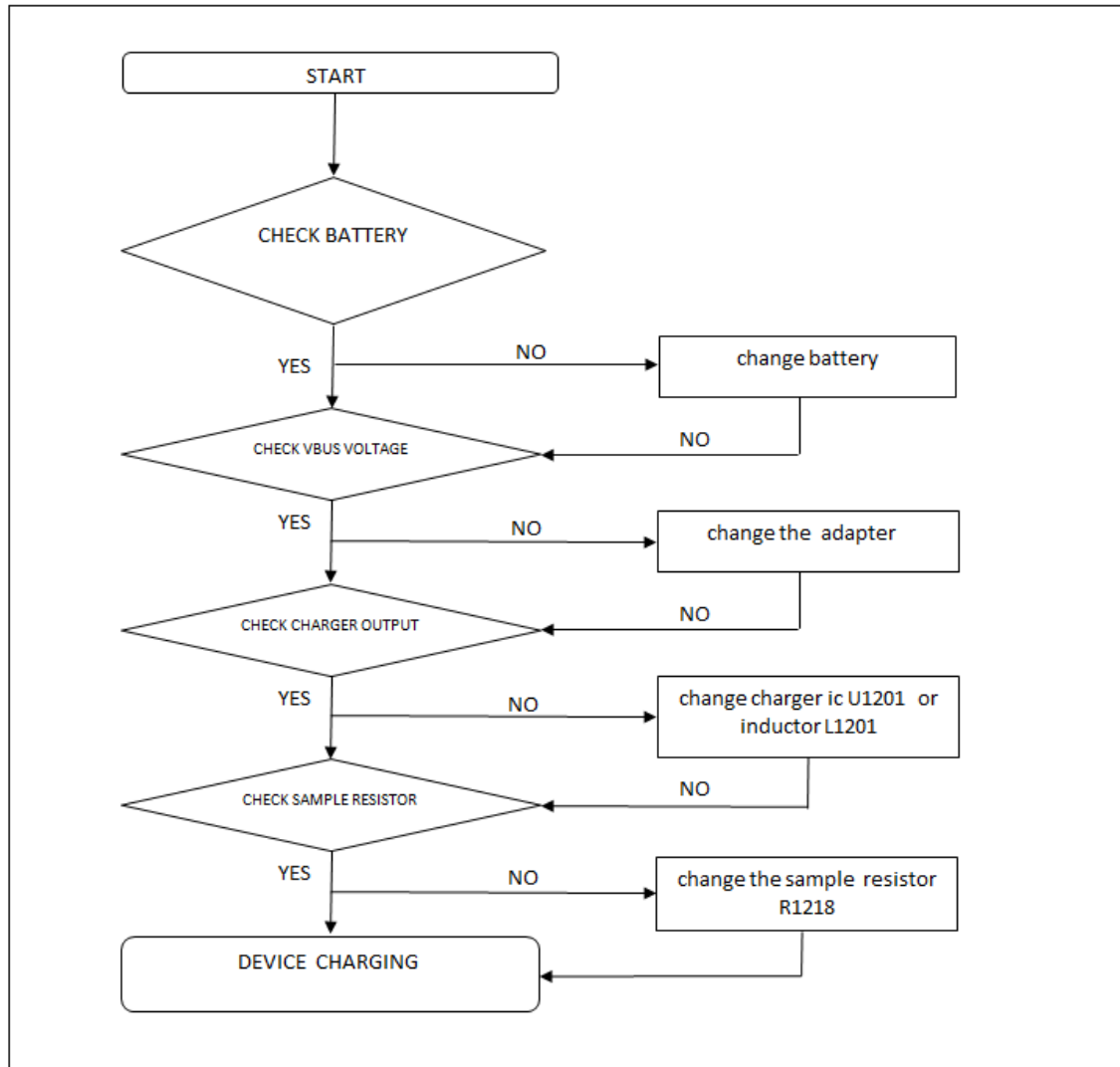
: Checking Power signal (LCM, PMU, Clock)



8. Level 3 Repair

8-4-2. Charging

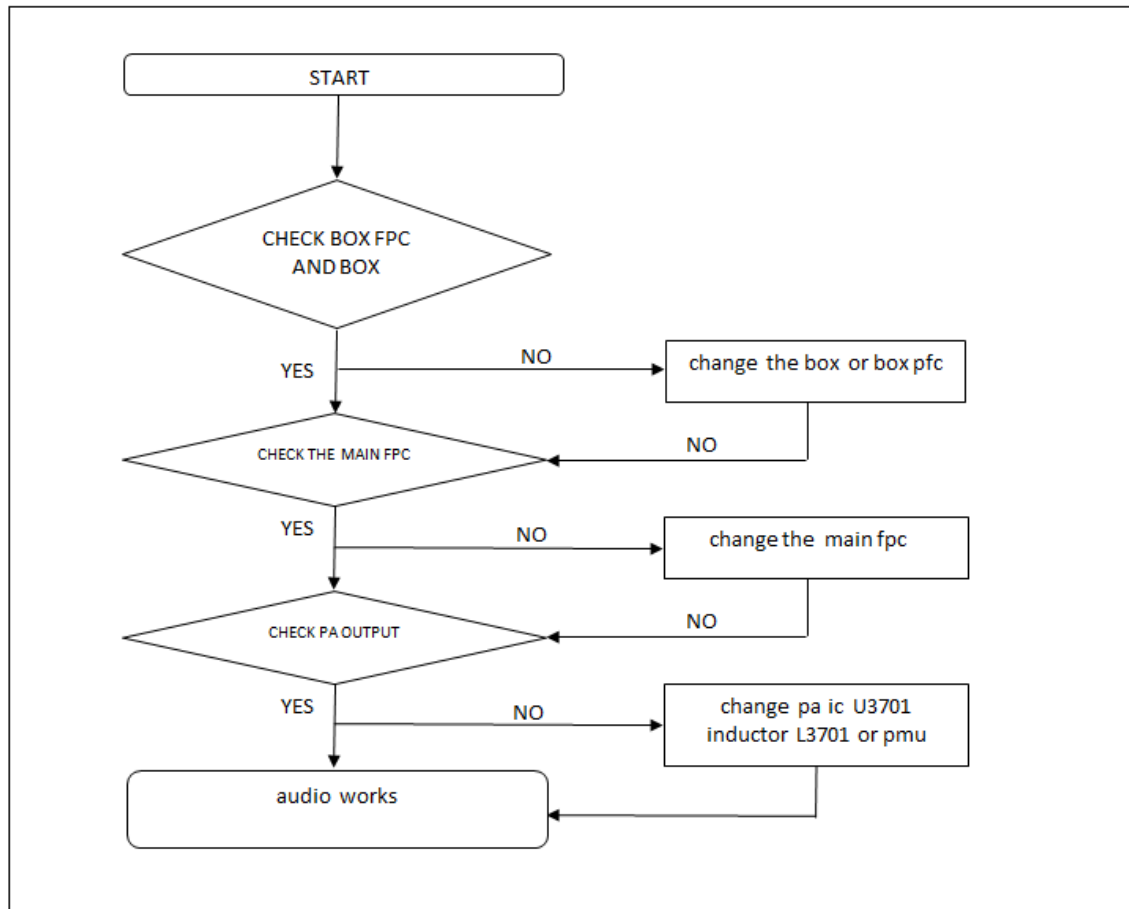
: The charging controlled by third charger chip



8. Level 3 Repair

8-4-3. Audio speaker

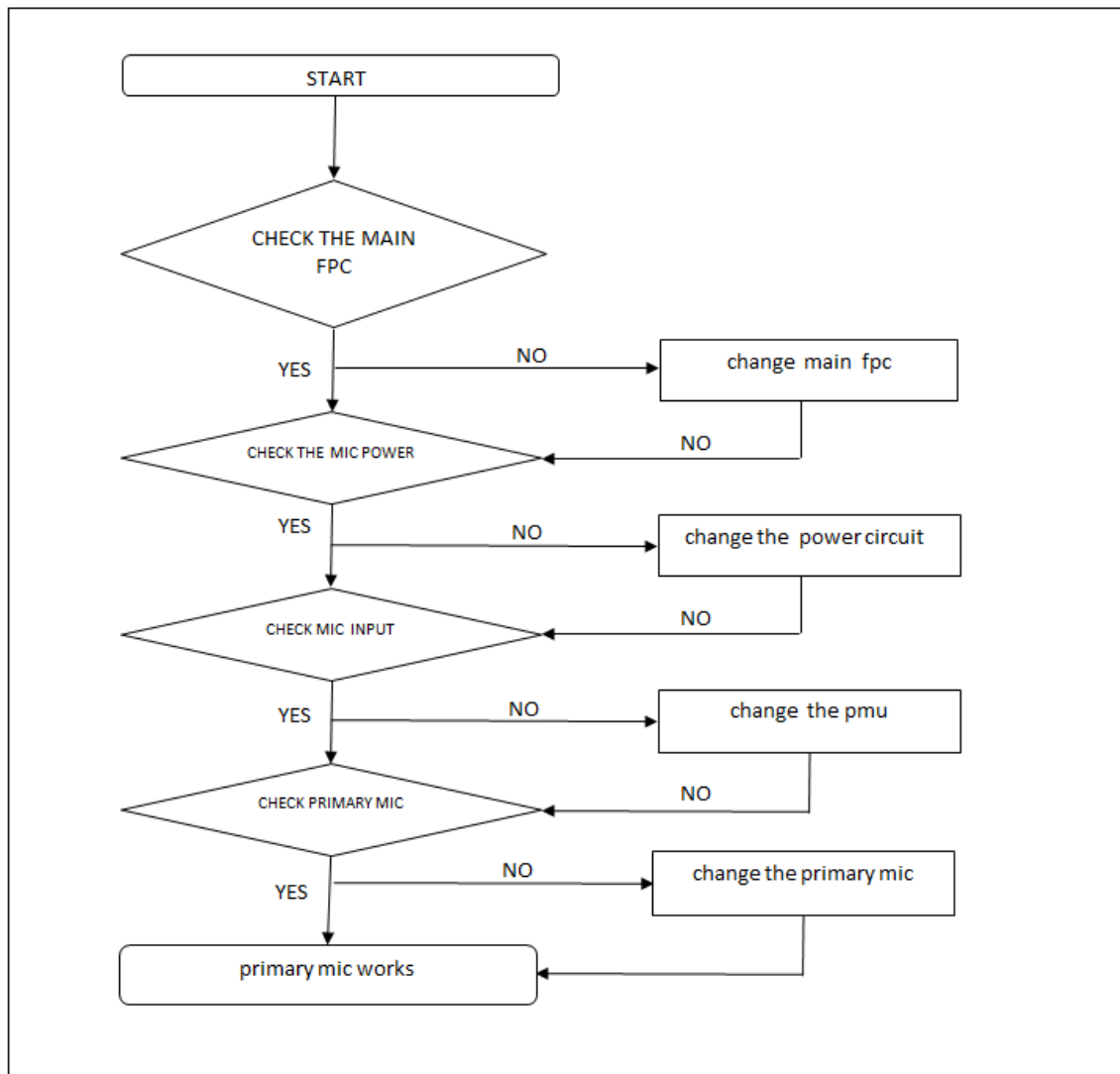
: The Speaker control signals are generated by MT6357 (U701) and PA AW87319 (U3701), the chip, PA and the speaker are to be checked out.



8. Level 3 Repair

8-4-4. Audio Main MIC

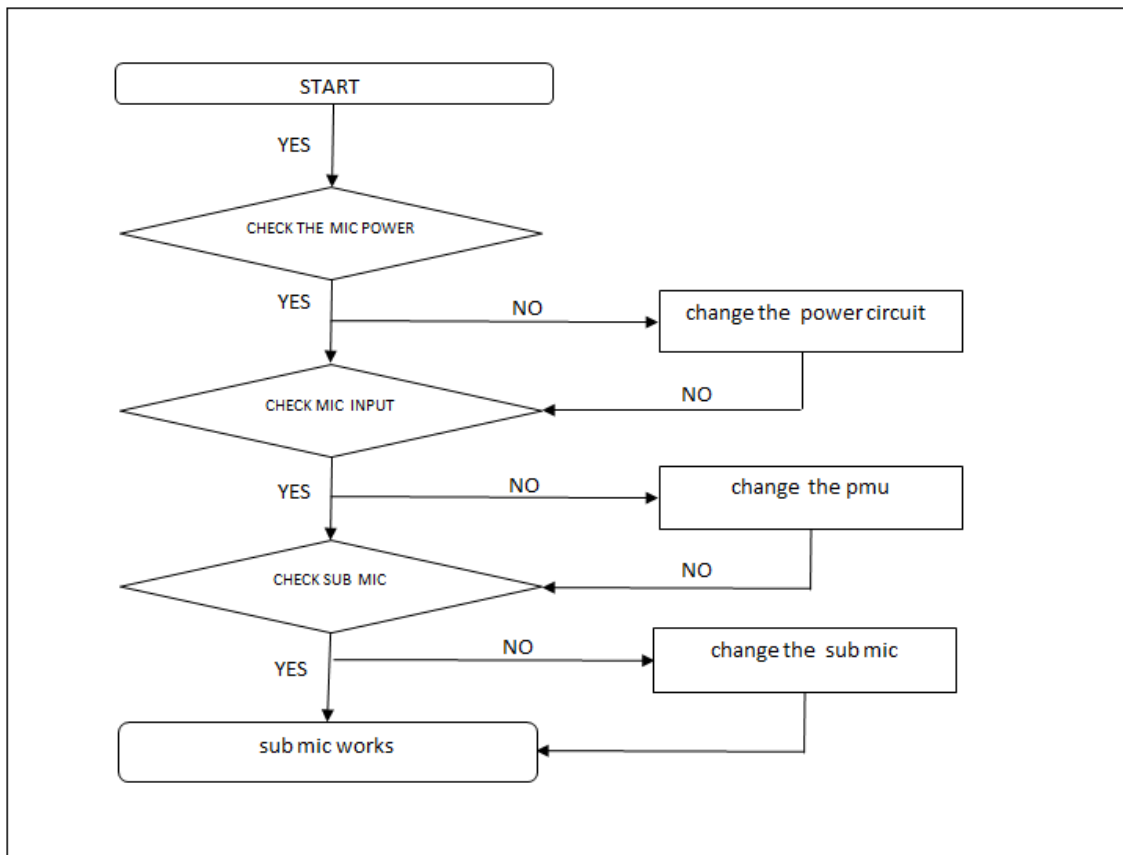
: The MIC control signals are generated by PMU chip MT6357 (U701), the PMU chip and the MIC are to be checked out.



8. Level 3 Repair

8-4-5. Sub MIC

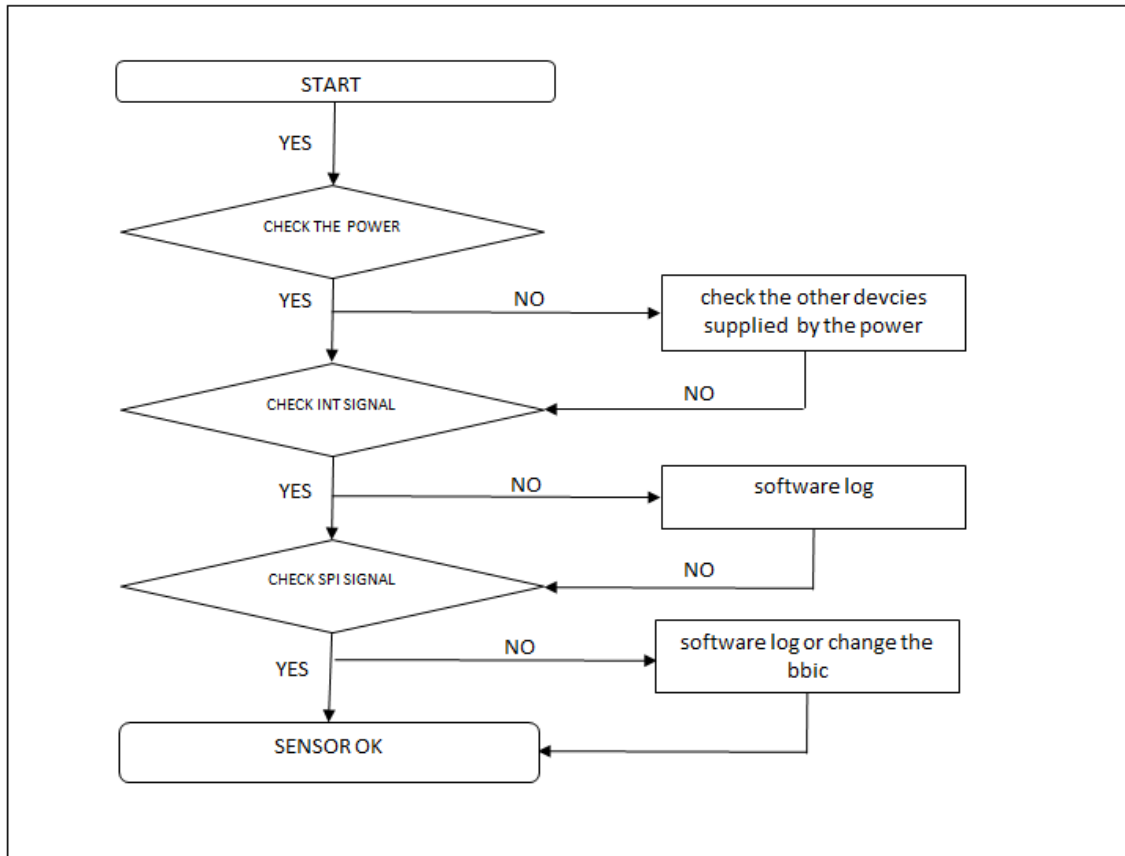
: The MIC control signals are generated by PMU chip MT6357 (U701), the PMU chip and the MIC are to be checked out.



8. Level 3 Repair

8-4-6. Accelerometer sensor

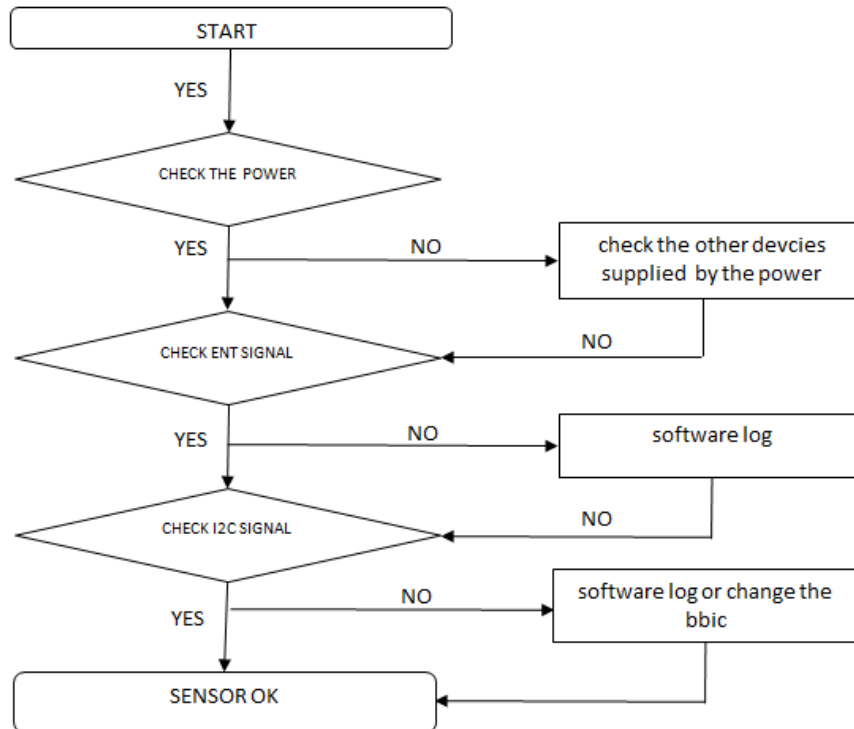
: The Accelerometer sensor is calibrated by using SW algorithm.



8. Level 3 Repair

8-4-7. Proximity and light sensor

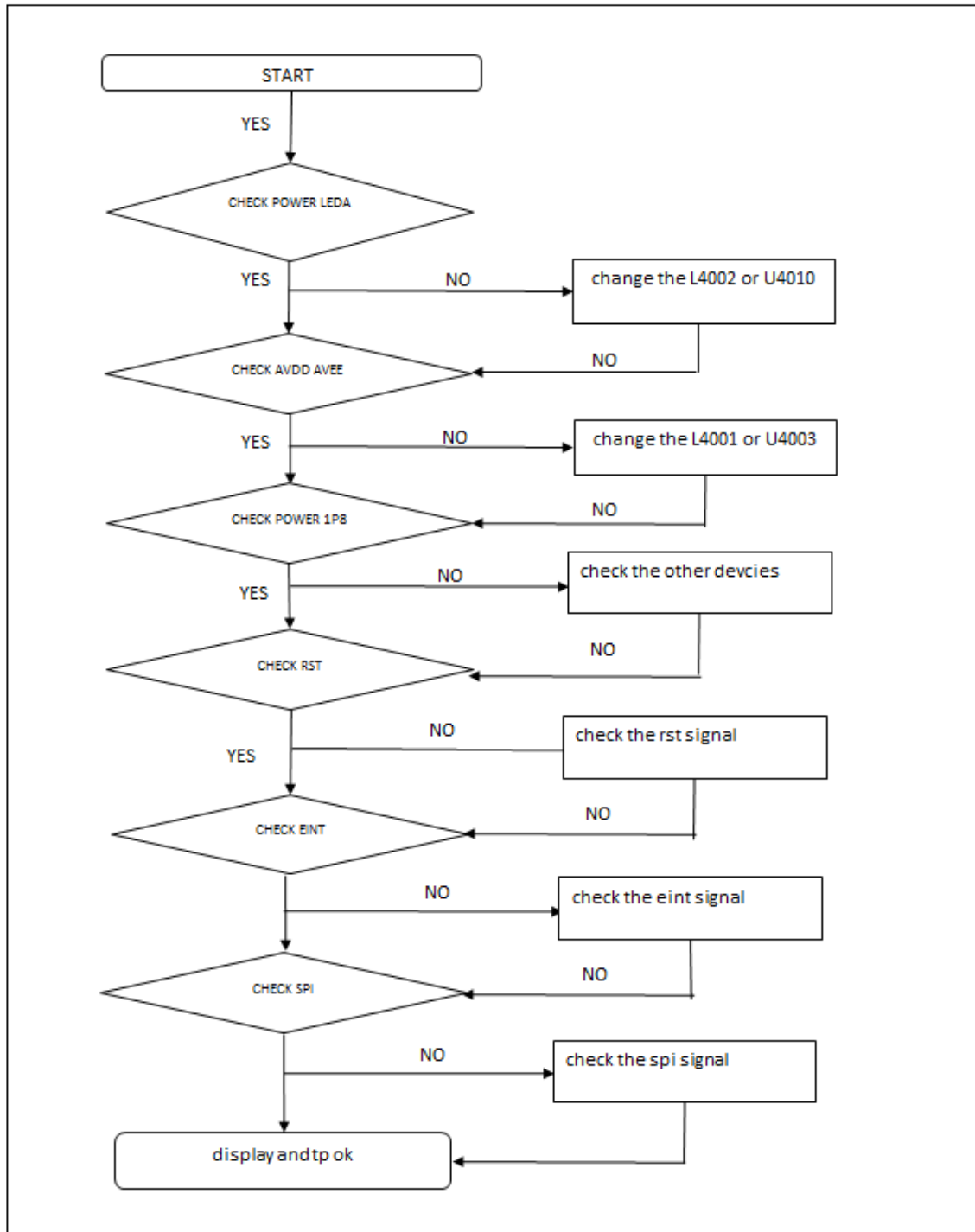
: Proximity (U4417) and Light Sensor (U4416) is worked as below: Control the screen's on/off operation automatically while making phone calls, and adjust the screen brightness according to ambient light.



8. Level 3 Repair

8-4-8. TOUCH SCREEN AND DISPLAY

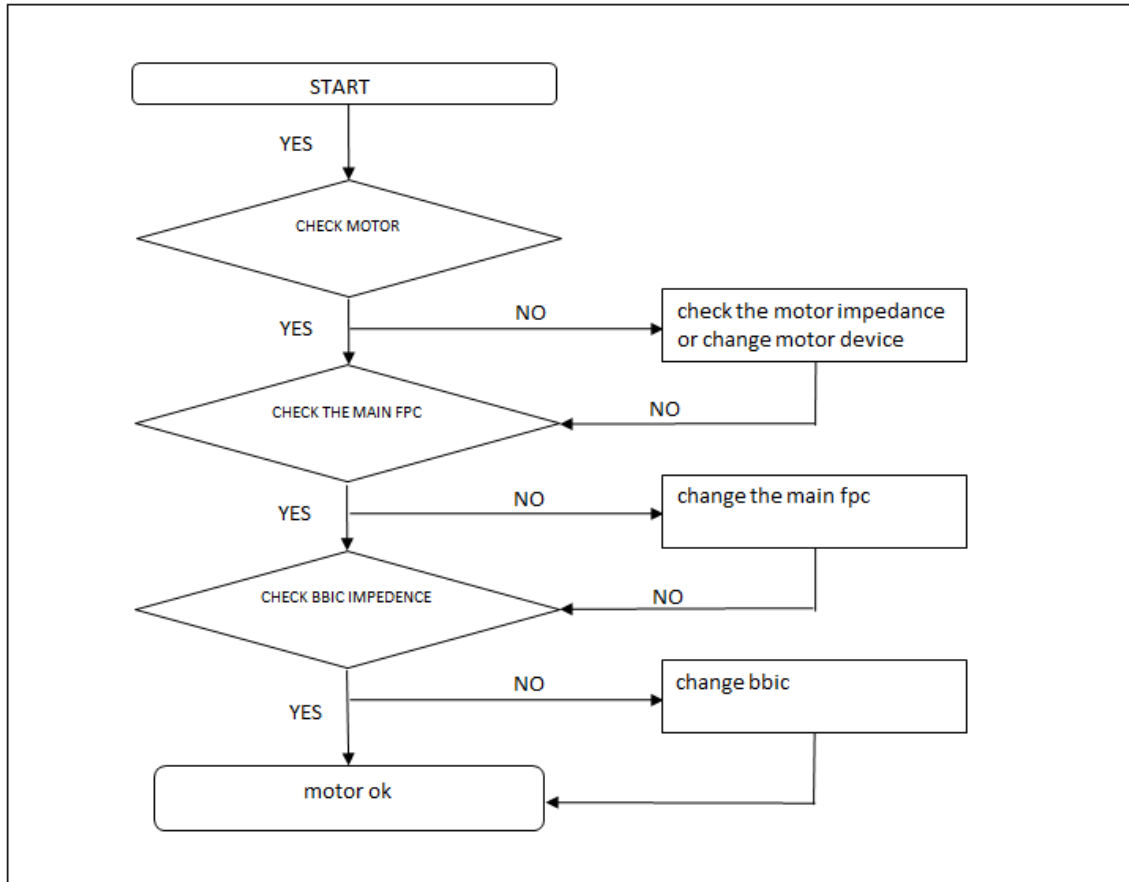
: The Touch control signals are generated by MT6762. It is assembled with LCD.



8. Level 3 Repair

8-4-9. Vibrator

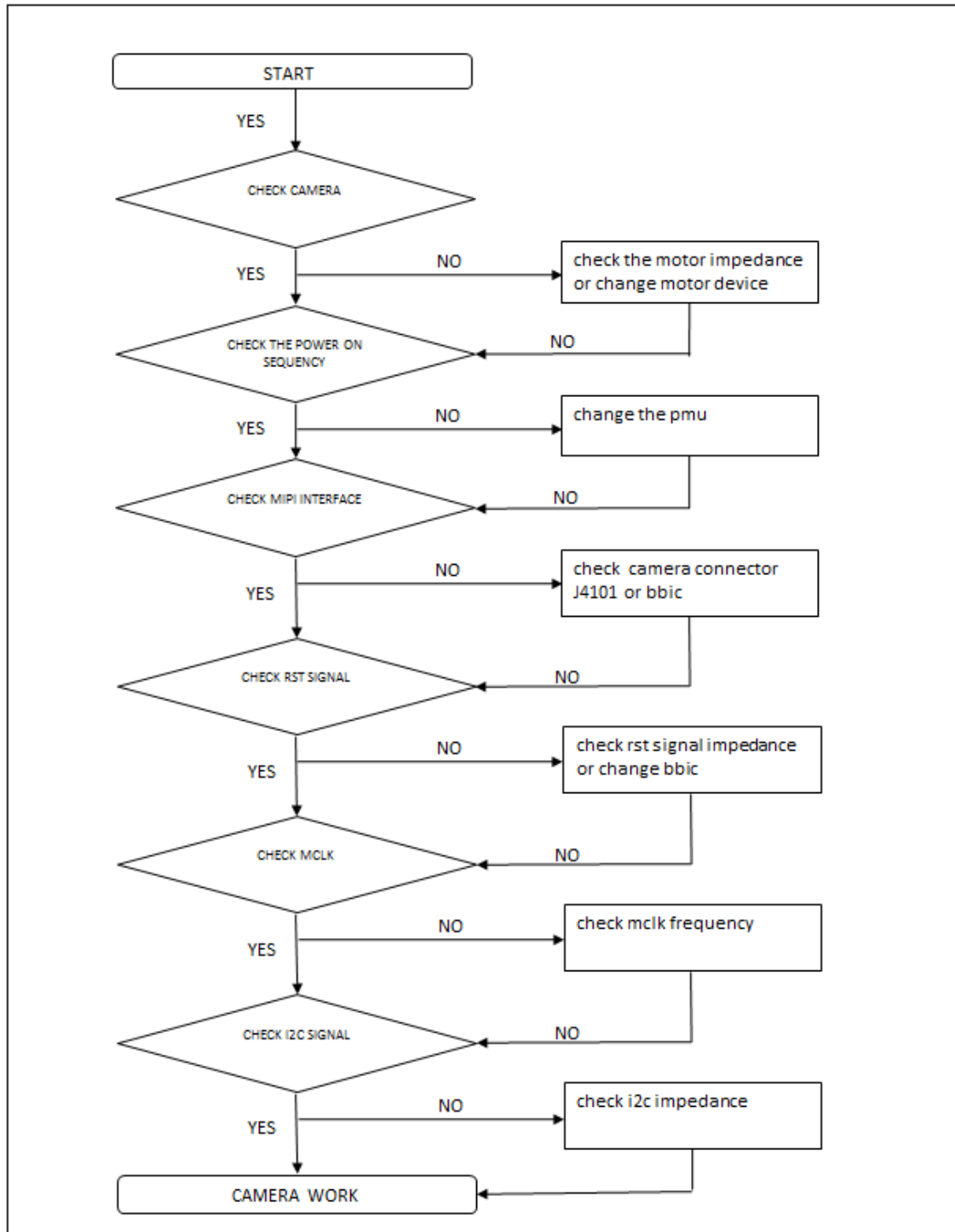
: The Vibrator control signals are generated by MT6357.



8. Level 3 Repair

8-4-10. Rear Camera

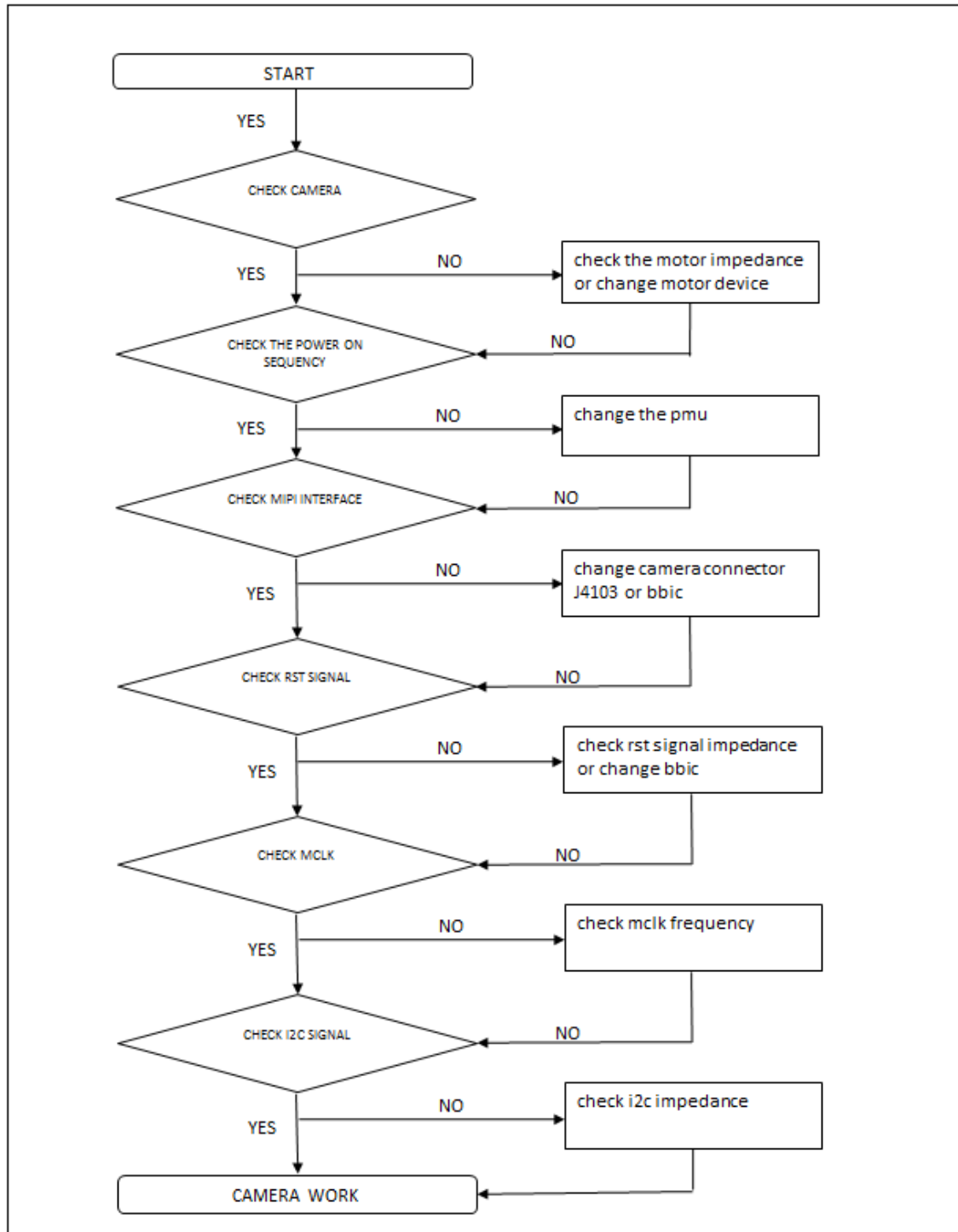
: The Camera control signals are generated by MT6762.



8. Level 3 Repair

8-4-11. Front Camera

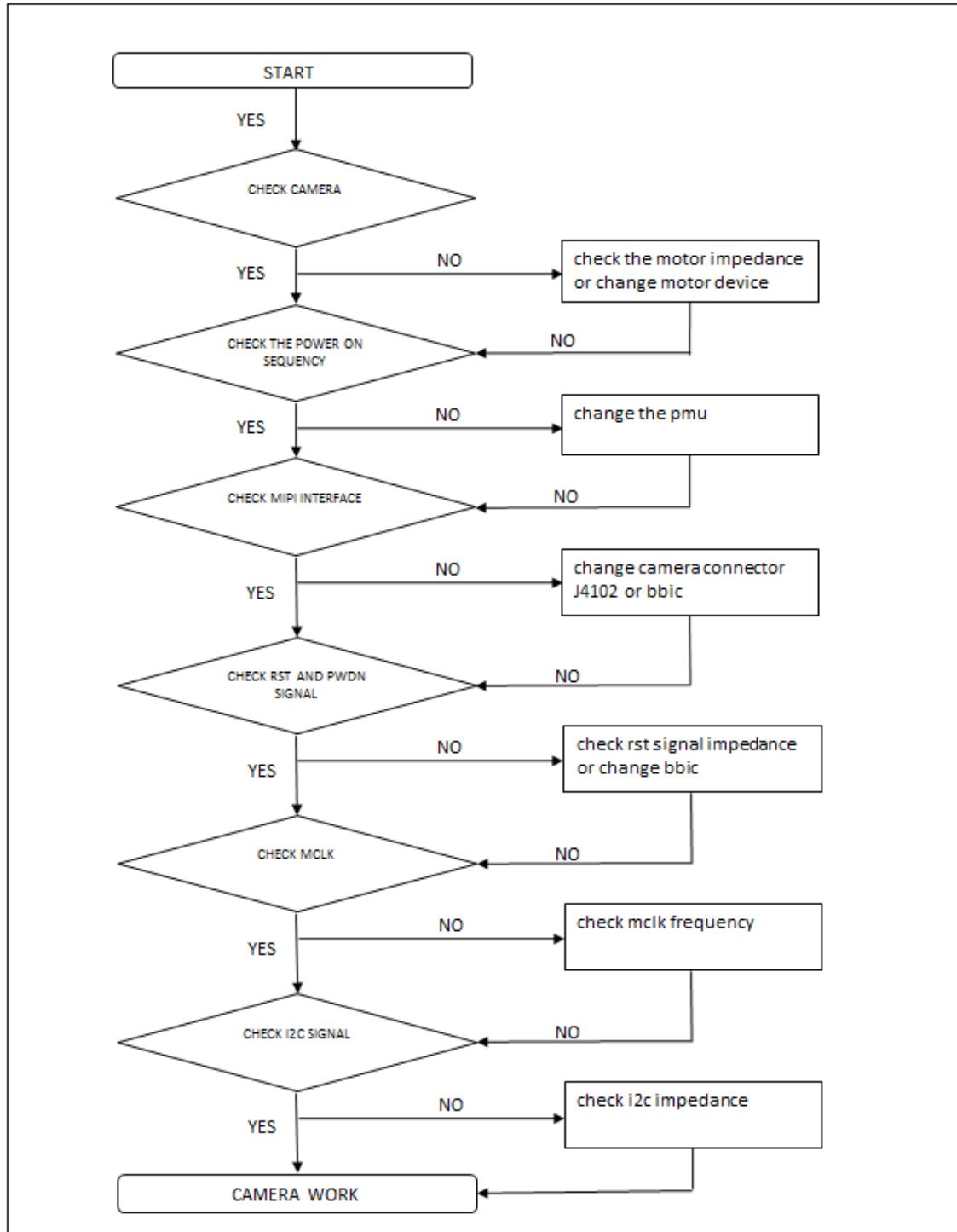
The Camera control signals are generated by MT6762.



8. Level 3 Repair

8-4-12. Depth Camera

: The depth Camera control signals are generated by MT6762.



8. Level 3 Repair

8-5. Service Schematics

- U101_MT6762_BB chip IC , Digital Baseband Processor(Top)

558	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28		
A	NC	WF_IN	WF_IP	EMIO_CAB			NC		EMIO_CAB								EMIO_DQ8		EMIO_DQ15		EMIO_DQ31		MSDC0_DAT4	MSDC0_DAT2	MSDC0_DAT7	DVDD1_8_MSD	NC	A		
B	WF_OP	WF_QN	EMIO_CAB	EMIO_CAB			EMIO_CAB	EMIO_CAB	NC			EMIO_DQ21	EMIO_DQ22	EMIO_DQ20		EMIO_DQ4	EMIO_DQ8	EMIO_DQ17		EMIO_DQ14	EMIO_DQ28	EMIO_DQ3		MSDC0_DAT5	MSDC0_DAT6	MSDC0_CMO	MSDC0_DAT1	USB_D_M	AVDD1_9_USB	B
C		DVSS	AVDD1_2_VDD	NC	EMIO_CAB	DVSS	EMIO_CAB		NC		EMIO_DQ18	EMIO_DQ17	DVSS	EMIO_DQ0	EMIO_DQ2	DVSS	EMIO_DQ1		EMIO_DQ13	EMIO_DQ14	EMIO_DQ28	EMIO_DQ3		MSDC0_DAT3	MSDC0_RSTB	MSDC0_DAT0	USB_D_P		C	
D	BT_IN	BT_IP	DVSS	NC	EMIO_CAB	EMIO_CAB	NC	EMIO_CAB	DVSS	EMIO_DQ16	NC	EMIO_DQ19	DVSS	EMIO_DQ30	EMIO_DQ5		EMIO_DQ31		EMIO_DQ12	EMIO_DQ7	EMIO_DQ27	DVSS	MSDC0_DAT1	MSDC0_CAB	CHD_3P	DVSS	AVDD1_8_USB	D		
E		BT_OP	DVSS	EMIO_CAB	EMIO_CAB	EMIO_CAB	EMIO_CAB	EMIO_CAB	EMIO_CAB	EMIO_DQ23		EMIO_DQ23	EMIO_DQ27	EMIO_DQ30	EMIO_DQ5		EMIO_DQ31		EMIO_DQ12	EMIO_DQ7	EMIO_DQ27	DVSS	NC	DVSS	DVSS	CHD_3P	SYSTRST_B	AVDD1_2_USB	E	
F	GPS_I	DVSS	BT_ON	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	EMIO_CAB		DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	EMIO_DQ10	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	TESTM_ODE		F
G	GPS_Q	CONN_HRST	CONN_HRST	DVSS			XIN_WBG				DVSS	DVSS	DVSS	DVSS	DVSS	NC						DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		G
H		AVDD1_8_WBG	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	AVDD1_8_WBG	AVDD1_8_WBG	AVDD1_8_WBG	AVDD1_8_WBG	DVSS	DVSS		AVDD1_8_WBG	AVDD1_8_WBG	AVDD1_8_WBG	AVDD1_8_WBG	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		H
J	DVDD1_8_WBG	CAM_C_LK2	SC16	SDA6	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST	CONN_HRST												DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		J
K		CAM_P_DN2	CAM_R_ST2	ANT_5_E12	ANT_5_E10	ANT_5_E11	CONN_TOP_D_ATA	CONN_TOP_D_ATA	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		K
L	CS1A_L1P	CS1A_L1N	CS1A_L1P	CS1A_L1N	CS1B_L1P	CS1B_L1N	DVSS	DVSS			DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		L
M	CS1A_L2P	CS1A_L2N	CS1B_L1P	CS1B_L1N	CS1B_L1P	CS1B_L1N	DVSS	DVSS			DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		M
N	CS1A_L1P	CS1A_L1N	CS1A_L1P	CS1A_L1N	CS1B_L1P	CS1B_L1N	DVSS	DVSS			DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		N
P	CS1A_L2P	CS1A_L2N	CS1B_L1P	CS1B_L1N	CS1B_L1P	CS1B_L1N	DVSS	DVSS			DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		P
R	CS2A_L1P	CS2A_L1N	CS2A_L1P	CS2A_L1N	CS2B_L1P	CS2B_L1N	DVSS	DVSS			DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		R
T	CS2A_L1P	CS2A_L1N	CS2A_L1P	CS2A_L1N	CS2B_L1P	CS2B_L1N	DVSS	DVSS			DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		T
U	DVDD1_8_WBG	AVDD1_2_VDD	CS2A_L1P	CS2A_L1N	CS2B_L1P	CS2B_L1N	DVSS	DVSS			DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		U
V	CAM_R_ST1	CAM_R_ST2	SDA2	SC12	SDA4						DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		V
W		CAM_C_LK1	ENT10	ENT9	CAM_C_LK1	CAM_C_LK1	ENT11	ENT11	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		W
Y	SRCLK_NAI	CAM_P_DN1	ENT7	ENT6	ENT4	ENT5	ENT5	ENT5	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		Y
AA	KPRO_W0	UTXD0	URXD0	ENT3	ENT2	ENT1	ENT1	ENT1	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		AA
AB	KPRO_W1	KPCOL_1	KPCOL_0	ENT0	SC10	CDMSP_5A					DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		AB
AC	SPI0_M_0	SPI0_M_1	PWM0	SDA0	CDMSP_5A	SPI0_M_0					DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		AC
AD	SPI0_M_1	SPI0_M_2	SPI0_M_3	SPI0_M_4	SPI0_M_5	SPI0_M_6					DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		AD
AE	SCL1	SPI0_M_1	SPI0_M_2	SPI0_M_3	SPI0_M_4	SPI0_M_5					DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		AE
AF	SDA1	SPI0_M_1	SPI0_M_2	SPI0_M_3	SPI0_M_4	SPI0_M_5					DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		AF
AG	NC	SPI0_M_1	SPI0_M_2	SPI0_M_3	SPI0_M_4	SPI0_M_5					DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	ADD_C_LK_MIS		AG
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28		

Figure 2-1. Ball map view